

Hacking, Phishing & Spoofing

Lesson 43 — Attacks and Deception

JKSSB Computer Awareness

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Why Attacks Succeed

- ▶ Many computer attacks do not break the machine; they **trick the person** using it.
- ▶ An attacker needs only two things: a **weakness** to use and a **person** who responds without checking.
- ▶ Weaknesses can be technical (weak password, old software) or human (trust, hurry, fear).
- ▶ This lesson teaches the names, the signs, and the **safe response** — not how to attack.

The one idea

Attacks use deception or a known weakness; defence is verification plus good habits.

Four Words to Keep Straight

- ▶ **Vulnerability**: a weakness in software, settings, or human habit that can be misused.
- ▶ **Exploit**: a method that takes advantage of a vulnerability.
- ▶ **Attack**: the complete harmful act against a person or system.
- ▶ **Payload**: the harmful part delivered by the attack, such as stolen data or a malicious file.

Read it as a chain

Vulnerability (weakness) → exploit (method) → attack (act) → payload (harm).

What Hacking Means

- ▶ **Hacking** means gaining **unauthorised access** to a computer, account, or network.
- ▶ A **hacker** is the person who does it; unauthorised access is unlawful even when no data is stolen.
- ▶ An **ethical hacker** tests security **with written permission** to find and fix weaknesses.
- ▶ Exam depth: hacking is the broad act; phishing and spoofing are specific deceptive methods inside it.

Trap alert

Hacking is unauthorised access. Permission separates an ethical security test from an attack.

Password Attacks

- ▶ **Guessing**: trying obvious passwords such as name, birth year, or password123.
- ▶ **Dictionary attack**: trying every word from a list of common passwords automatically.
- ▶ **Brute-force attack**: trying **every possible combination** until one works; longer passwords resist it.
- ▶ **Shoulder surfing**: watching the keyboard or screen while the victim types.

Keep the pair straight

Dictionary tries a word list; brute force tries every combination.

Strong Password Habits

- ▶ Use a **long** password with mixed letters, digits, and symbols; never reuse it across accounts.
- ▶ Never share a password or an **OTP** (One-Time Password) with anyone, including a caller who claims to be bank staff.
- ▶ Lock the screen when leaving the desk; cover the keypad while typing a PIN.
- ▶ Use **two-step verification** where offered: password plus a code on the phone.

Office desktop example

A clerk who uses the same short password for email and office login loses both accounts from one guess; a long unique password with two-step verification blocks the same attempt.

Social Engineering

- ▶ **Social engineering** means tricking a person into giving access, instead of breaking software.
- ▶ **Pretexting**: inventing a false story (“bank officer”, “IT support”) to extract details.
- ▶ **Baiting**: offering something free (prize, free recharge, free software) to trigger a click or download.
- ▶ Pressure words — **urgent**, **blocked**, **verify now** — are the standard signal.

Safe response

Slow down, verify through a known channel, and never disclose a password or OTP under pressure.

Phishing

- ▶ **Phishing** is a fake message, mail, or call that **imitates a trusted sender** to steal passwords, OTPs, or money.
- ▶ The message carries a **malicious link** to a look-alike login page, or a **malicious attachment** to open.
- ▶ Common channels: email, **SMS smishing**, and voice-call **vishing**.
- ▶ Signs: urgent tone, spelling errors, unknown sender, and a link that does **not** match the real website address.

Definition

Phishing steals credentials or money by imitation plus a link, attachment, or call to action.

Spoofing

- ▶ **Spoofing** means **faking an identity**: the sender address, caller number, or website looks genuine but is forged.
- ▶ **Email spoofing**: the “From” address shows a trusted name that was never the real sender.
- ▶ **Caller ID spoofing**: the displayed phone number is forged to look like a bank or office.
- ▶ **Website (URL) spoofing**: a look-alike address (bank-verlfy.com for bank-verify.com) hosts a fake login page.

What spoofing is

A forged identity used as a disguise; phishing often **uses** spoofing to look trustworthy.

Phishing vs Spoofing vs Hacking

Term	What it is	One-line test
Hacking	Unauthorised access	Was access taken without permission?
Phishing	Fraud to steal data/money	Was a fake message asking for action?
Spoofing	Forged identity	Was the sender or address faked?

Trap alert

A phishing mail **uses** a spoofed address; spoofing is the mask, phishing is the fraud, hacking is the broad act.

Malicious Links

- ▶ A **malicious link** leads to a fake login page or starts an unwanted download.
- ▶ Hover over the link to read the **real address** before clicking; on a phone, long-press to preview it.
- ▶ Type the website address yourself instead of tapping a link in an unexpected message.
- ▶ Shortened or misspelled addresses in an urgent message are a warning sign.

Office desktop example

A mail says the salary slip is ready and points to `offlce-portal.com`; the real portal is `office-portal.com`. Typing the known address avoids the fake page.

Malicious Attachments

- ▶ A **malicious attachment** is a file that harms the computer or steals data when opened.
- ▶ Do not open an unexpected attachment, even from a known name — confirm with the sender first.
- ▶ Be cautious with files that ask to “enable macros” or “enable content” on opening.
- ▶ Exam depth: attachment is the file; link is the address; both can carry the payload.

Safe habit

Unexpected file → stop, verify the sender, delete if unverified.

Safe Habits That Block Most Attacks

- ▶ Verify the sender through a **known channel** before acting on any urgent request.
- ▶ Keep software and **antivirus** updated; scan external drives before opening files.
- ▶ Back up important files so one incident cannot destroy all copies.
- ▶ Report phishing messages using the mail client's **report phishing** option instead of forwarding or replying.

If You Are Targeted

- ▶ Do **not** enter a password, OTP, or card details on a page reached from a suspicious message.
- ▶ Disconnect doubt: close the page, delete the message, and do not call back numbers given in it.
- ▶ Change the password from the **real website**, and inform the bank or office through its published helpline.
- ▶ Keep a record (screenshot, sender address) for the complaint.

Order of action

Stop → verify on the real channel → change credentials → report.

Trap Alert: Attack Facts to Keep Straight

- ▶ Phishing **imitates** a trusted sender to steal data; spoofing **forges** the identity it wears.
- ▶ OTP stands for **One-Time Password**; no genuine bank or office asks for it by call or message.
- ▶ Dictionary tries a **word list**; brute force tries **every combination**.
- ▶ Vulnerability is the **weakness**; exploit is the **method**; payload is the **harm delivered**.
- ▶ Ethical hacking needs **written permission**; without it, access is unauthorised.

Lesson 43: Recall

- ▶ Hacking is unauthorised access; ethical hacking has written permission.
- ▶ Social engineering tricks the person; pretexting invents a story and baiting offers a lure.
- ▶ Phishing is fake-message fraud for credentials or money; spoofing is the forged identity it may wear.
- ▶ Password attacks include guessing, dictionary, brute force, and shoulder surfing; long unique passwords plus two-step verification resist them.
- ▶ Malicious links and attachments carry the payload; verify the sender, type the address, and never share an OTP.
- ▶ Vulnerability → exploit → attack → payload is the chain to remember.